Connecting Devices

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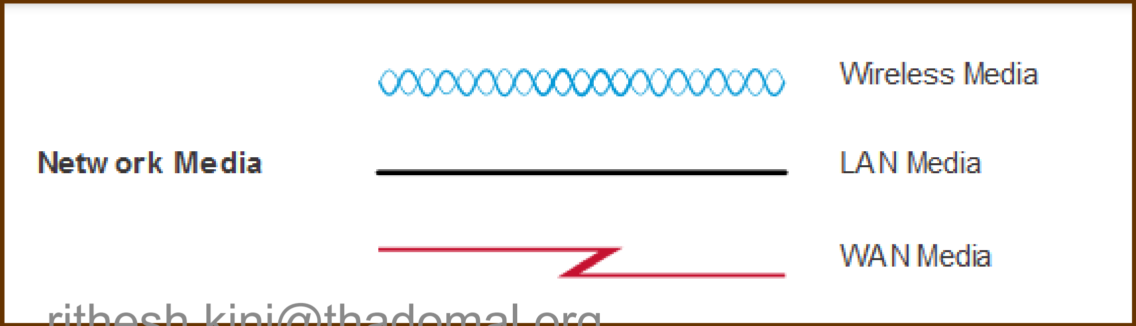
Network Components

- The network infrastructure contains three categories of hardware components:

- End devices
- Intermediate (or Connecting) devices

- Network media

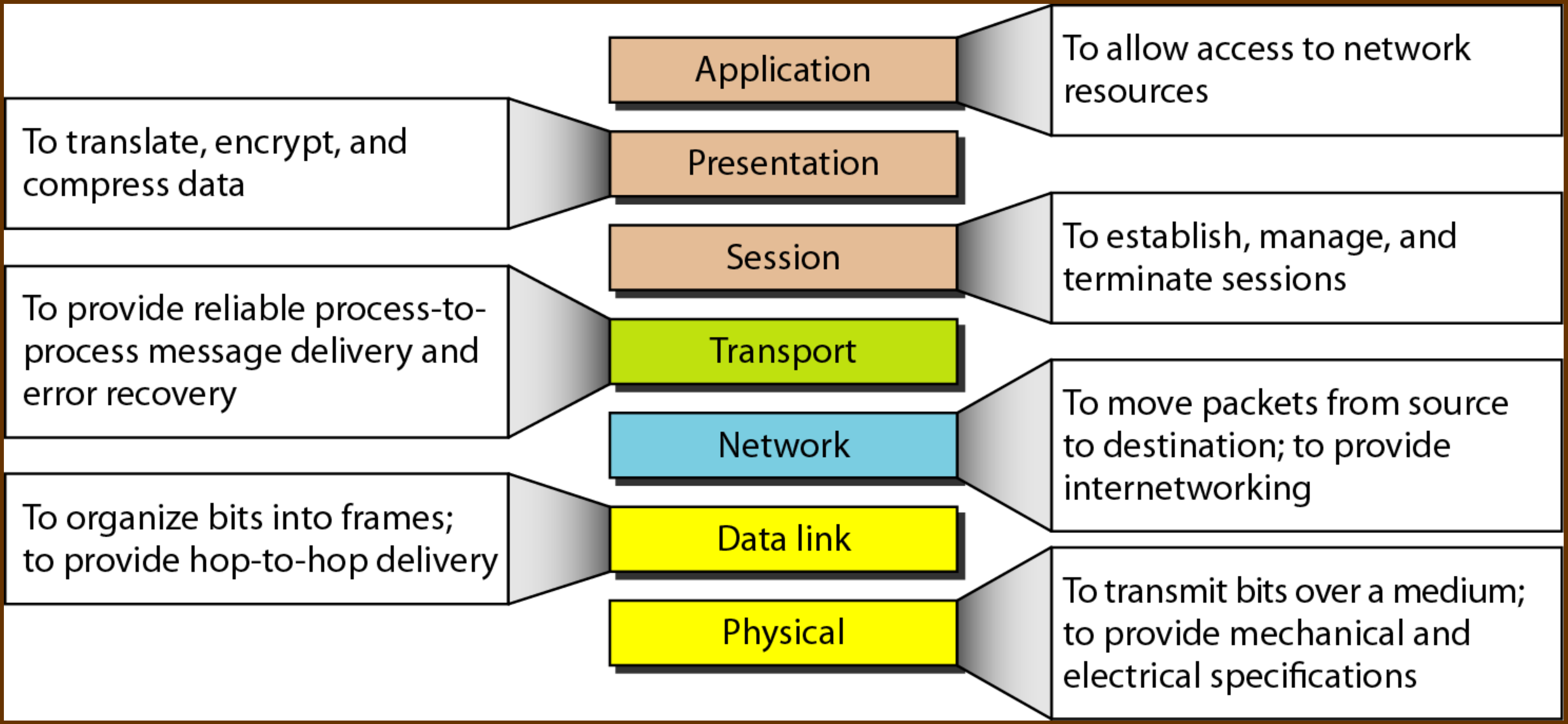


Intermediate (or Connecting) devices

- **Connecting devices :** are used to connect hosts together to make a network or to connect networks together to make an internetwork(internet).

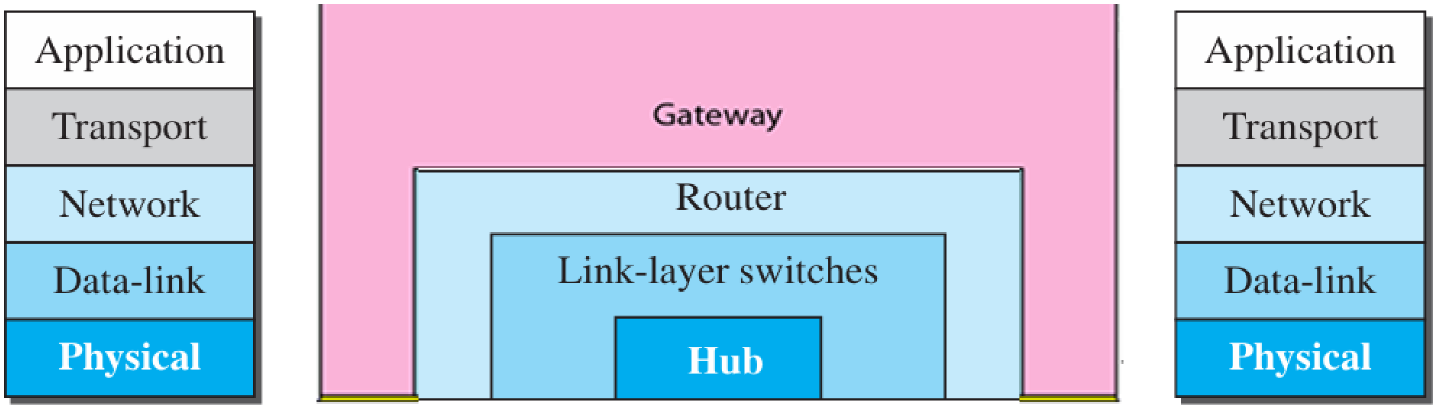
- Some of the common connecting devices are:
  - Repeaters
  - Hubs
  - Switches
  - Routers
  - Gateways

Summary of layers of OSI Model



Intermediate (or Connecting) devices

Figure 17.1 *Three categories of connecting devices*

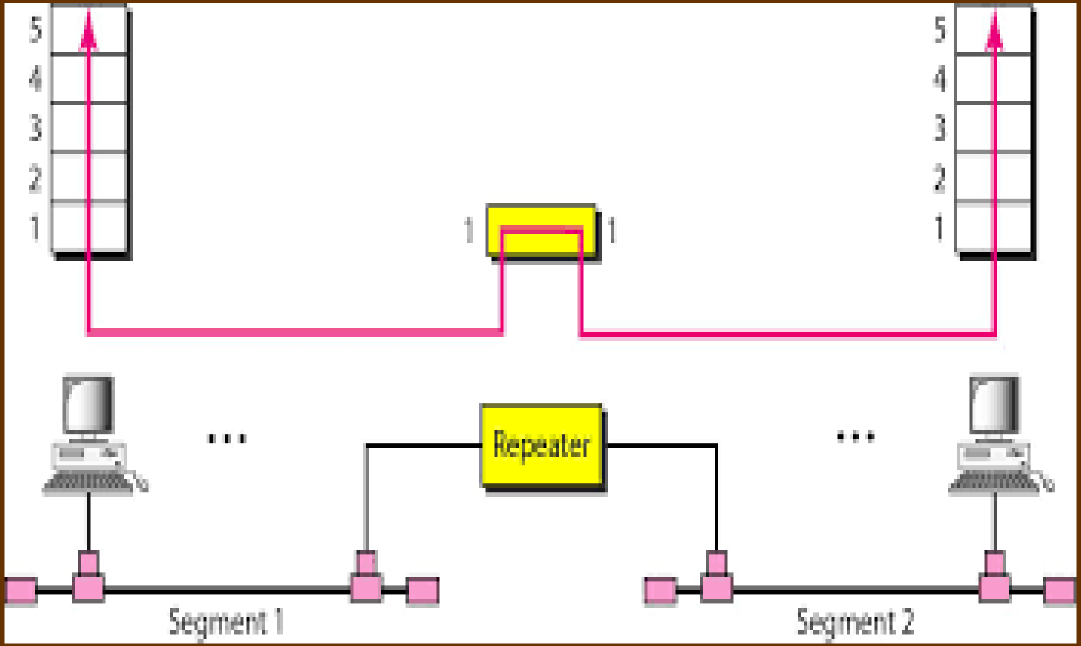


Intermediate (or Connecting) devices

1. Repeaters

- Its job is to **regenerate** the signal over the **same network** before the signal becomes too weak or corrupted.
- It helps in extending the length to which the signal can be transmitted over the **same network**.
- An important point to be noted about repeaters is that they do not amplify the signal.
- It is used in **Physical Layer**

Intermediate (or Connecting) devices



Repeater Device

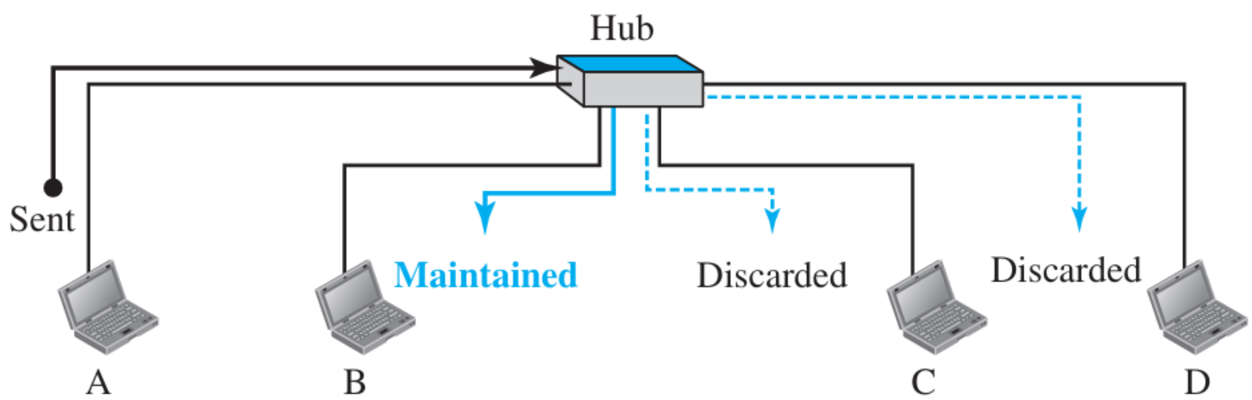
Intermediate (or Connecting) devices

2. Hub

- Like the repeater a **hub** is a device that operates only in the **physical layer**.
- In the past, when Ethernet LANs were using bus topology, a repeater was used to connect two segments of a LAN to overcome the length restriction of the coaxial cable.
- Today, however, Ethernet LANs use **star topology**. In a star topology, a repeater is a multiport device, often called a hub, that can be used to serve as the connecting point and at the same time function as a repeater.

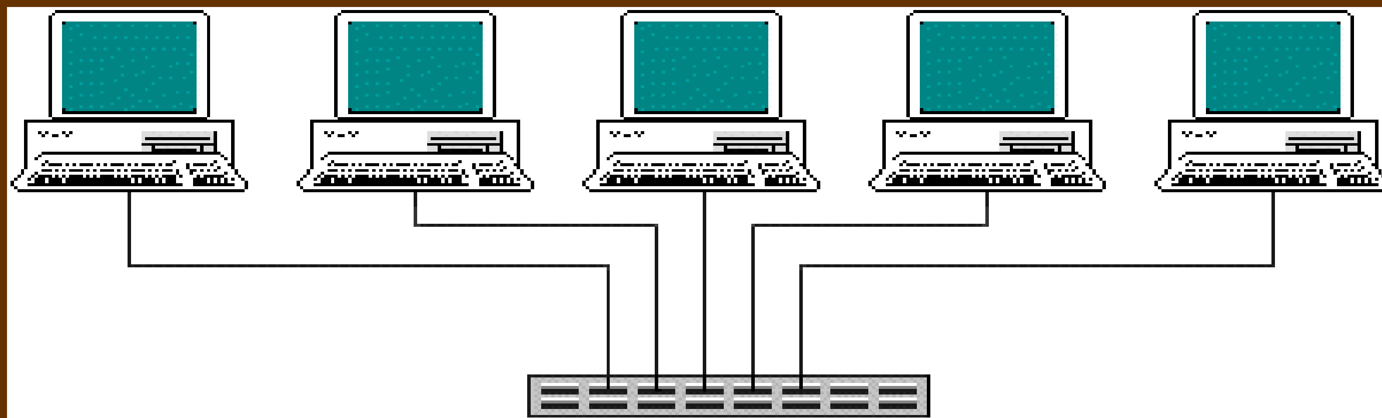
Intermediate (or Connecting) devices

Figure 17.2 *A hub*



- **Station A** sends a packet for **station B** that arrives at the hub. The hub broadcasts the frame. All stations in the LAN receive the frame, but only station B keeps it and others discard.

Intermediate (or Connecting) devices



Hub Device

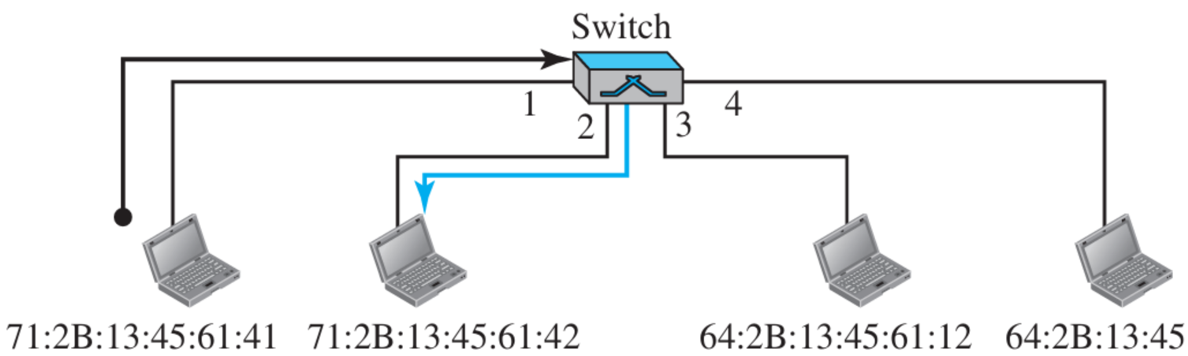
Intermediate (or Connecting) devices

3. Switch

- Like Hub **Switch** is a multiport device.
- Switch operates in both the **physical and the data-link layers**.
- As a physical-layer device, it regenerates the signal it receives.
- As a link-layer device, the switch can check the MAC addresses (source and destination) contained in the frame.
- A switch has filtering capability. It can check the destination address of a frame and can decide from which outgoing port the frame should be sent (next slide).
- Switches can perform error checking before forwarding data.
- They are considered very efficient by not forwarding packets that have errors or forwarding good packets selectively to correct devices only.

Intermediate (or Connecting) devices

Figure 17.3 *Link-layer switch* - Filtering Capability



Switching table

Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3
64:2B:13:45:61:13	4

- If a frame destined for station 71:2B:13:45:61:42 arrives at port 1, the switch consults its table to find the departing port. According to its table, frames for 71:2B:13:45:61:42 should be sent out only through port 2; therefore, there is no need for forwarding the frame through other ports.

Intermediate (or Connecting) devices

Switch

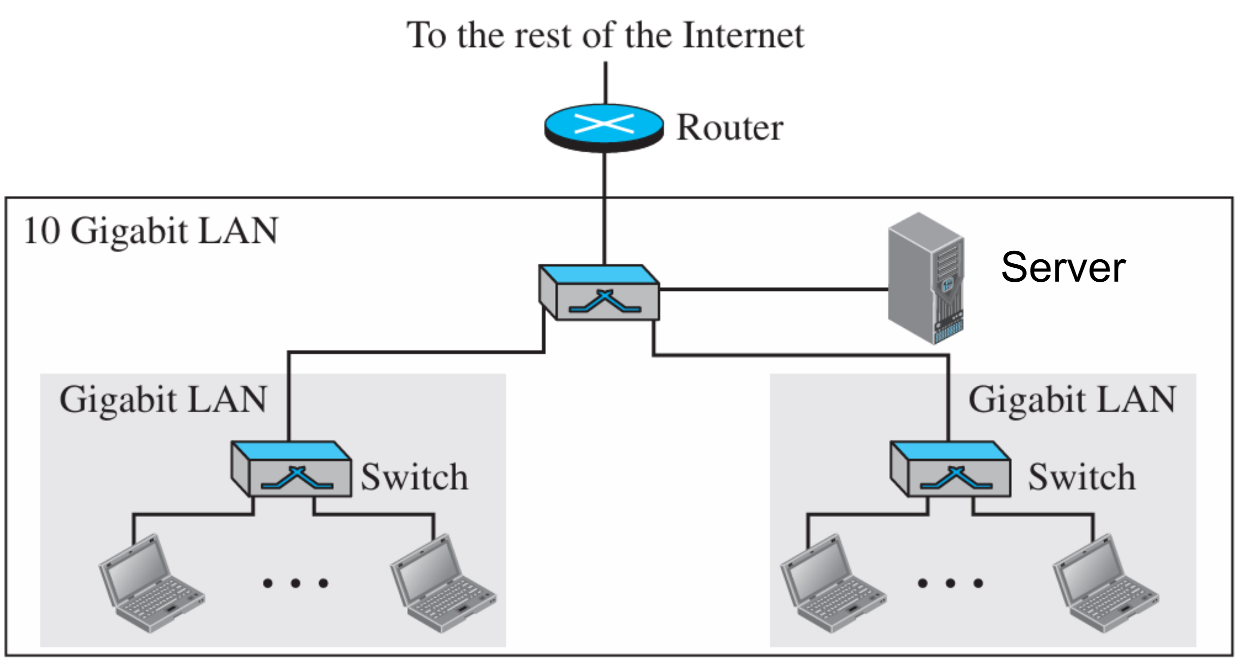
Intermediate (Connecting) devices

4. Router

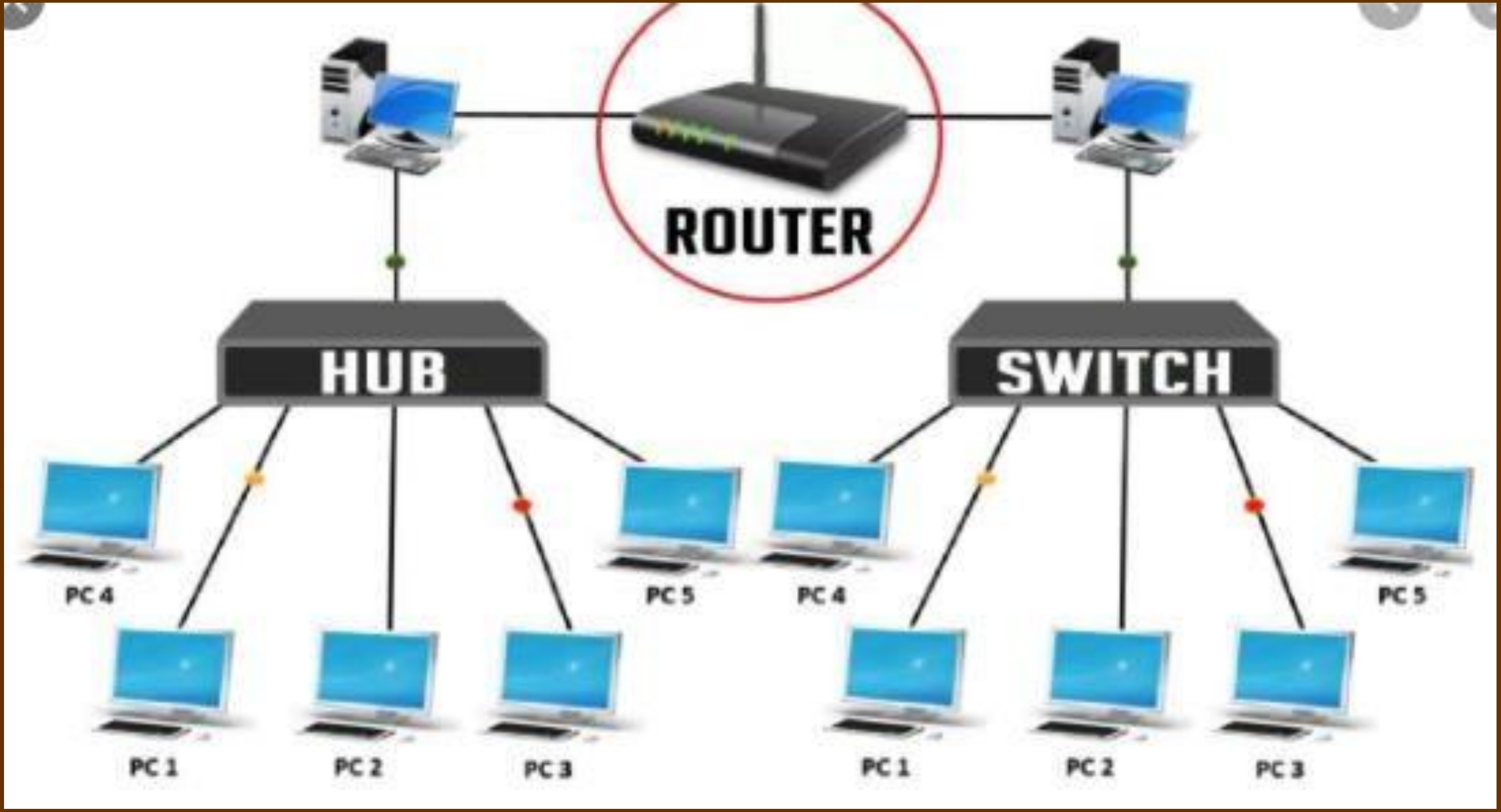
- A **router** is a <u>**three-layer device**</u>; it operates in the <u>physical, data-link, and network layers</u>.
- As a physical-layer device, it regenerates the signal it receives. As a link-layer device, the router checks the physical addresses (source and destination) contained in the packet. As a network-layer device, a router checks the network-layer addresses (IP addresses).
- A router is an internetworking device; it connects independent networks to form an internetwork. Routers normally connect LANs and WANs together.
- They have a dynamically updating routing table based on which they make decisions on routing the data packets.

Intermediate (or Connecting) devices

Figure 17.9 *Routing example*



Intermediate (or Connecting) devices



Intermediate (or Connecting) devices

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5. Gateways

- A Gateway is a single component that enables the data flow from one network to other
- It act as routers or switches that are capable of interacting with multiple networks that use dissimilar communication protocols.
- It can work on seven layers of the OSI.
- The default gateway is a router that connects your host to remote network segments.
- It can be implemented completely in hardware, software, or as a combination of both.

Connecting devices

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Repeater	Hub	Switch	Router	Gateway
Physical Layer	Physical Layer	Data Link Layer	Network Layer	ALL 7 layers

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